

Lab 8: Monte Carlo Integration Methods

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1 Introduction

The objective of this laboratory was the empirical verification of Monte Carlo methods for numerical integration. The exercise focuses on evaluating the integral of three functions on the interval $(0, 1)$:

1. $f_1(x) = x^2 + x + 1$
2. $f_2(x) = \sqrt{1 - x^2}$
3. $f_3(x) = \frac{1}{\sqrt{x}}$

The study compares a custom implementation of the basic “Hit-and-Miss” algorithm against advanced routines provided by the GNU Scientific Library (GSL): PLAIN, MISER, and VEGAS [1]. Theory indicates that for standard Monte Carlo methods, the error estimate decreases proportionally to $1/\sqrt{N}$, where N is the number of sample points.

2 Methodology

To perform the measurements, a test program was implemented in C using GSL library functions: `gsl_monte_plain_integrate`, `gsl_monte_miser_integrate`, and `gsl_monte_vegas_integrate`.

The research procedure included the following considerations:

1. **Hit-and-Miss Method:** This basic approach requires defining a finite bounding box. It works correctly for bounded functions (f_1 and f_2) but fails for $f_3(x) = 1/\sqrt{x}$, because the function approaches infinity as $x \rightarrow 0$, making it impossible to establish a maximum boundary y_{max} .
2. **GSL Integration:** The GSL routines were used to integrate $f_3(x)$. These algorithms are designed such that random sample points are always chosen strictly within the integration region, automatically avoiding endpoint singularities like the one at $x = 0$.

3 Results

3.1 Hit-and-Miss Convergence

The execution errors for the Hit-and-Miss algorithm applied to functions f_1 and f_2 are presented in Table 1.

Figure 1 illustrates the convergence in a logarithmic scale, showing the expected steady decrease in error as the number of trials increases.

Table 1: Absolute integration errors for the Hit-and-Miss method for $f_1(x)$ and $f_2(x)$.

N	Error $f_1(x)$	Error $f_2(x)$
100	0.213333	0.0546018
500	0.0926667	0.00860184
1000	0.0603333	0.00160184
5000	0.00986667	0.00100184
10000	0.0174333	0.000101837
50000	0.00729333	0.00402184
100000	0.00236667	0.00213184
500000	0.00165333	0.000432163

3.2 GSL Methods Comparison

Table 2 and Figure 2 present the error for the $f_3(x)$ integral using three distinct GSL strategies.

Table 2: Integration errors for $f_3(x) = 1/\sqrt{x}$ using GSL PLAIN, MISER, and VEGAS algorithms.

N	Error PLAIN	Error MISER	Error VEGAS
100	0.1281730	0.5453990	0.00140105
500	0.0395817	0.0152057	0.00323800
1000	0.3370740	0.0488994	0.00170075
5000	0.0011285	0.0135611	0.00006517
10000	0.0166513	0.0300152	0.00003832
50000	0.0121750	0.0016634	0.00002639
100000	0.0131327	0.0098215	0.00001162
500000	0.0005657	0.0001767	0.00000216

The results show clear disparities based on the variance reduction technique used: * **PLAIN**: Relies on pure random sampling and yields the highest error at $N = 500000$ (5.65×10^{-4}). * **MISER**: Uses recursive stratified sampling [3] to concentrate points in regions of highest variance. It improves upon the PLAIN method, achieving an error of 1.76×10^{-4} . * **VEGAS**: Based on importance sampling [2], sampling from the probability distribution described by $|f|$. It heavily outperforms the others, achieving an error of 2.15×10^{-6} for the same number of function calls.

4 Conclusion

The conducted experiments confirm the theoretical principles of Monte Carlo integration:

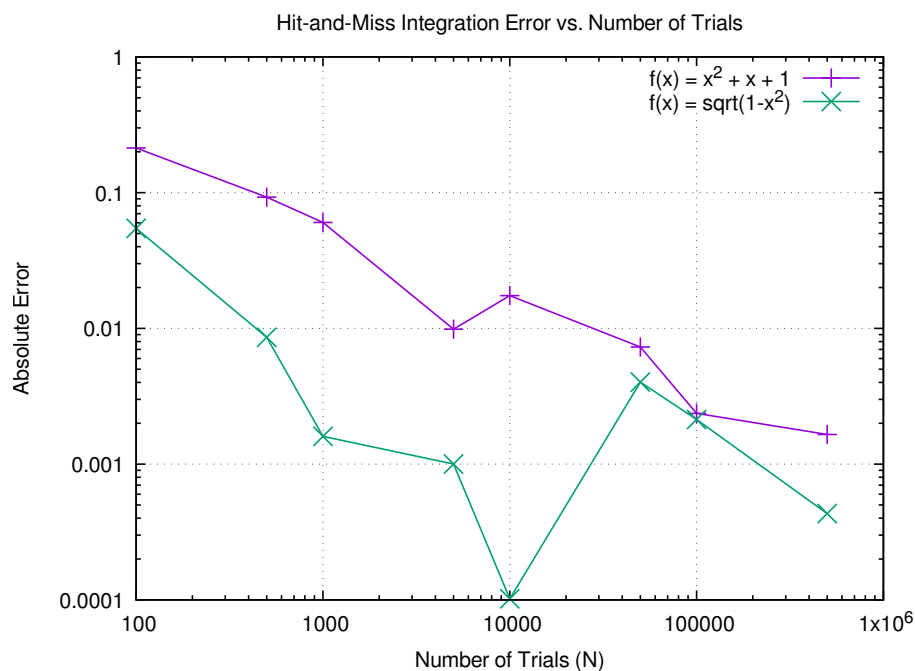


Figure 1: Hit-and-Miss Integration Error vs. Number of Trials. The downward trend is consistent with standard Monte Carlo error scaling.

1. The rudimentary Hit-and-Miss method is effective only for bounded functions.
2. The GSL routines handle endpoint singularities gracefully by sampling strictly within the bounds.
3. Variance reduction techniques are critical for efficiency. The VEGAS algorithm, utilizing importance sampling, proved to be highly superior for an integrand with a sharp localized peak ($1/\sqrt{x}$), requiring significantly fewer iterations to reach a high degree of precision compared to uniform random sampling.

Bibliography

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- [3] Press, W.H., Farrar, G.R.: Recursive stratified sampling for multidimensional monte carlo integration. *Computers in Physics* **4**(2), 190–195 (1990)

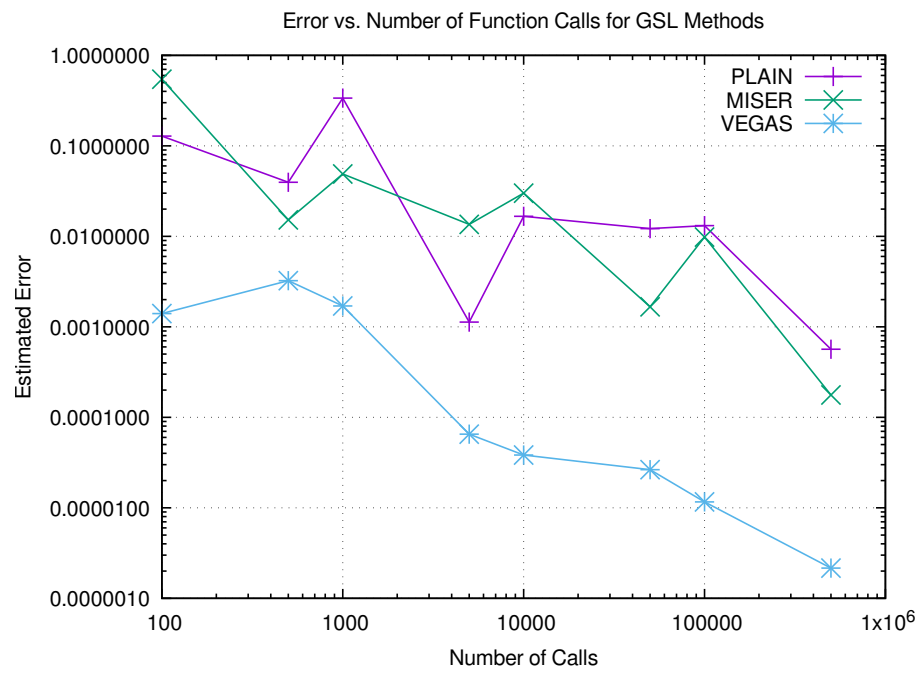


Figure 2: Error vs. Number of Function Calls for GSL Methods. Note the significant accuracy advantage of the VEGAS algorithm.